Q14 1) Exothermic

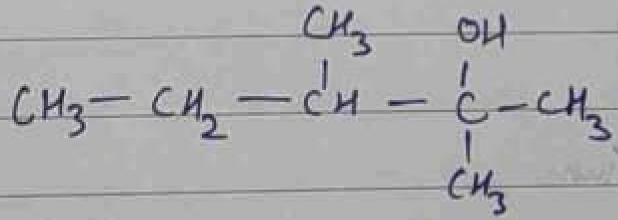
Q15 1) CO_2

Q16 2) 2

Q17 4) ii & iii)

Q18 3) i-c, ii-d, iii-b, iv-a

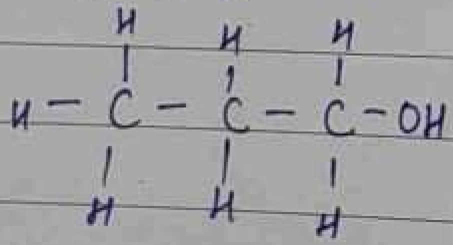
Q19 2)



Q20 2) 148

• Lateral Inversion

Q25 Propanol, C_3H_7OH



Q26 i) ~~Aluminium~~ Magnesium

ii) Copper

Q32 Metal → Hg (Mercury)

Ore → HgS (Cinnabar)

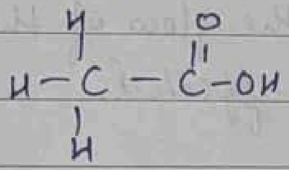
~~$HgS \xrightarrow{\Delta}$~~

$$2HgS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2HgO_{(s)} + 2SO_{2(g)}$$ (Roasting)

$$2HgO_{(s)} \xrightarrow{\Delta} 2Hg_{(l)} + O_{2(g)}$$ (Direct heating)

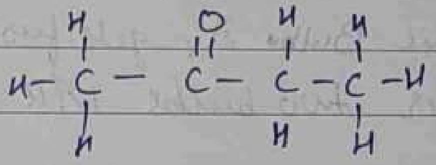
Q33 i) Ethanoic acid

CH_3COOH



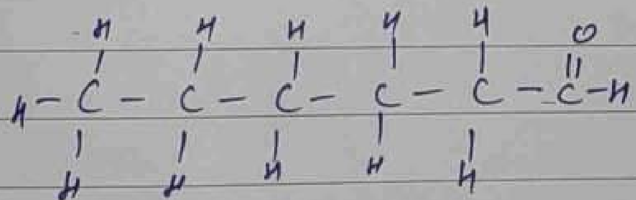
ii) Butanone

C_4H_8O



iii) Hexanal

$C_6H_{12}O$



Q36 a) i) $2FeSO_{4(s)} \xrightarrow{\Delta} Fe_2O_{3(s)} + SO_{3(g)} + SO_{2(g)}$

ii) X → SO_2 (Sulphur dioxide)

iii) Thermal decomposition

b) Zinc is more reactive than hydrogen, hence it displaces it to form $ZnCl_2$ but copper is less reactive than hydrogen and hence can't displace it.

Q37 i)

$$\cancel{H}\,\underbrace{2NaCl + 2H_2O}_{\text{Brine}} \xrightarrow{\text{electricity}} 2NaOH_{(s)} + Cl_{2(g)} + H_{2(g)}$$

ii) Anode → Chlorine

Cathode → Hydrogen

iii) Chlorine → Disinfecting water

Hydrogen → Fuel in rockets

iv) $NaCl + H_2O + NH_3 + CO_2 \rightarrow \underset{\text{(baking soda)}}{NaHCO_3} + NH_4Cl$